MODULE 6A · EPISODE 2

Light Phenomena and Applications

Where Theory Meets Everyday Observation

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Light Phenomena and Applications

This episode picks up where Episode 1's look at the fundamental nature of light left off, and asks what happens when light meets matter. The panel works through reflection, refraction, and diffraction, using Huygens' Principle and the Principle of Superposition to explain them; total internal reflection and the fiber optics it makes possible; how our eyes turn light into color; the constant speed of light and why light slows down in water and glass; how light lets astronomers look back in time; and the centuries-long effort to measure light's speed, from Galileo's lanterns to laser pulses bounced off the Moon.

The Panel

Name	Role
Dr. Edward Thorne	Host; astrophysicist
Dr. Lena Petrova	Guest; exoplanet researcher
Dr. Kenji Tanaka	Guest; microbial ecologist

Learning Objectives

After this episode you should be able to:

- Distinguish specular from diffuse reflection and state the Law of Reflection.
- Explain refraction as a change in light's speed at a boundary between optical media of different optical densities, and predict which way a ray bends.
- Use Huygens' Principle and the Principle of Superposition to explain refraction and diffraction.
- Describe Young's double-slit experiment and explain bright and dark bands in terms of path difference.
- Explain total internal reflection and the critical angle, and how optical fibers use them.
- Describe how rods and the three kinds of cones produce our vision, and use additive color theory to build white, yellow, cyan, and magenta.
- Define the index of refraction and use it to find the speed of light in a medium.
- Explain Cherenkov radiation without contradicting the constancy of c.
- Explain why looking farther out in space means looking back in time, and trace the history of measuring the speed of light.

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QUICK REFERENCE

Key Terms

Term	Definition
Specular reflection	Mirror-like reflection from a smooth surface, in which image integrity is maintained.
Image integrity	The ability to see an object in a reflection and still identify it; distortion (as in a curved mirror) is still allowed.
Diffuse reflection	Reflection in random directions from a dull, rough surface; no image is formed.
Albedo	Reflectivity; the fraction of incoming light a surface reflects.
Normal	The line perpendicular to a surface at the point where a ray strikes it.
Law of Reflection	The angle of incidence equals the angle of reflection, both measured from the normal; the incident ray, reflected ray, and normal lie in the same plane.
Refraction	The bending of light as it crosses a boundary between optical media of different optical densities, because its speed changes.
Optical density	A measure of how much a transparent medium slows light; light travels more slowly in an optically denser medium.
Huygens' Principle	Each point on a wave front acts as a point source of circular wavelets.
Principle of Superposition	Overlapping waves add together; used with Huygens' Principle to find how a wave front advances.
Snell's Law	The mathematical law describing how much a ray bends when it is refracted.
Diffraction	The bending of light around corners and obstacles, and its spreading out as it passes through narrow openings, with no change of medium.
Coherent light	Light whose waves stay in step, such as laser light; ideal for demonstrating diffraction.
Interference	The combining of overlapping waves: constructive where they reinforce, destructive where they cancel.
Maxima and minima	The bright bands (constructive interference) and dark bands (destructive interference) in an interference pattern.
Path difference	The extra distance light from one slit travels compared with light from the other slit.
Total internal reflection	Complete reflection of a ray back into a denser medium when it strikes the boundary beyond the critical angle.
Critical angle	The angle of incidence in the denser medium beyond which every ray is totally internally reflected.
Optical fiber	A thin strand of very pure glass or plastic, with a core and a lower-density cladding, that guides light by total internal reflection.

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Term	Definition
Rods	Retinal cells responsible for black-and-white (gray-scale) vision in dim light.
Cones	Retinal color cells; red, green, and blue cones (long-, medium-, and short-wavelength) respond to overlapping ranges of light.
Additive color	Making colors by adding red, green, and blue light; all three at full intensity make white, none makes black.
Complementary colors	Two colors that add to make white: yellow and blue, cyan and red, magenta and green.
c	The speed of light in a vacuum, about 3×10^8 m/s; the same for every observer.
Index of refraction	The speed of light in a vacuum divided by the speed of light in a medium: $n = c \div v$.
Cherenkov radiation	Blue light given off by charged particles moving faster than light travels in a medium, but slower than c.
Stellar aberration	The slight shift in the apparent positions of stars caused by Earth's motion around the Sun.
Lunar Laser Ranging	Timing laser pulses reflected from the Moon to measure its distance.
Retroreflector	An array of corner-cube prisms that sends light straight back toward its source.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
Angle in = angle out	Law of Reflection: angle of incidence equals angle of reflection
90°	Largest possible angle of refraction; Snell's Law predicting more means total internal reflection
About 1801	Thomas Young demonstrates the wave nature of light
Whole number of λ	Path difference that gives a bright band (maximum)
Odd number of λ/2	Path difference that gives a dark band (minimum)
3	Kinds of cones in the human retina: red, green, and blue
About 565 nm	Peak sensitivity of red (long-wavelength) cones
About 535 nm	Peak sensitivity of green (medium-wavelength) cones
About 430 nm	Peak sensitivity of blue (short-wavelength) cones
About 400 to 700 nm	Visible spectrum, from the violet end to the red end
About 3×10^8 m/s	c, the speed of light in a vacuum
299,792.458 km/s	Current known value of c

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Value	What it refers to
About 186,000 mi/s	c in miles per second
About 1.3 seconds	Light-travel time from the Moon to Earth
About 7.5 times	Trips around Earth in one second at the speed of light
About 500 s (8.3 min)	Light-travel time from the Sun to Earth
1.33	Index of refraction of water; light travels at about three-fourths of c in water
More than 13 billion years	Age of the oldest galaxy light seen by the Hubble Space Telescope
About 13.5 billion years	Galaxies seen by the James Webb Space Telescope, a few hundred million years after the Big Bang
1676	Ole Rømer times eclipses of Io; estimate about 220,000 km/s
About 1728	James Bradley uses stellar aberration; about 301,000 km/s
1849	Hippolyte Fizeau, toothed wheel; about 313,000 km/s in air
1862	Léon Foucault, rotating mirror; about 298,000 km/s
1879 and 1926	Albert Michelson; about 299,910 and 299,796 km/s
5	Long-used lunar retroreflectors: Apollo 11, 14, and 15, and two Lunokhod rovers
About 3.8 cm per year	Rate at which the Moon is moving away from Earth

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — Reflection

Mirrors are very smooth and highly reflective, with a high **albedo**: they reflect most of the light that strikes them. They show **specular reflection**, in which **image integrity** is maintained: you can see an object in the reflection and still identify it. Image integrity does not rule out distortion. Flat mirrors reflect without distortion, but curved mirrors can enlarge, shrink, or warp the image.

A dull, rough surface produces **diffuse reflection**, scattering light in random directions. Either way, every ray obeys the same **Law of Reflection**: the angle between the incident ray and the **normal** (the line perpendicular to the surface where the ray strikes) equals the angle between the reflected ray and that normal. Put simply, **the angle of incidence equals the angle of reflection**, and the incident ray, reflected ray, and normal all lie in the same plane.

That one law explains both kinds of reflection. On a mirror, parallel rays from an object all meet the surface with the same normal, so they are all reflected at the same angle and the image holds together. On a microscopically rough surface, the normal points a different way at every tiny spot, so the same parallel rays scatter at many different angles.

Part 2 — Refraction

We know from experiment that **refraction** occurs because light changes speed as it crosses a boundary between two optical materials of different **optical densities**. Wave theory explains it with **Huygens' Principle**, which treats each point on a wave front as a point source of circular wavelets. Combined with the **Principle of Superposition**, it shows that a plane wave front stays a plane wave front in the second medium, but must bend, because light changes speed when it enters.

Going from a less dense optical medium into a denser one, light slows down, and the ray bends **in toward the normal**. Going into a medium of lower optical density, it bends **away from the normal**. **Snell's Law of Refraction** describes this mathematically.

Refraction is everywhere. The corneas and lenses of healthy eyes refract light to form **real, inverted, and reduced** images in focus on the retinas, the light-sensitive screens at the back of our eyes. And a pencil half immersed in a glass of water appears broken at the surface, an optical illusion caused by refraction.

Part 3 — Diffraction and Young's Experiment

Diffraction is the bending of light around corners and obstacles, and its spreading out as it passes through narrow openings. Unlike refraction, there is **no change of medium**. Huygens' Principle and the Principle of Superposition explain it too. If light were made of tiny classical particles traveling only in straight lines, as Isaac Newton believed, it could not diffract. Diffraction happens for all light, visible or invisible, single-frequency or mixed, but **coherent** visible laser light is ideal for showing it.

Around **1801**, the Englishman **Thomas Young** convincingly demonstrated light's wave nature. He sent a plane wave front of light through an aperture plate with two narrow, closely spaced

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slits. The light spreading out from the slits overlapped and produced an **interference pattern** on a distant screen: a row of alternating bright and dark bands. The bright bands, or **maxima**, are where the wavelets from the two slits interfere constructively; the dark bands, or **minima**, are where they interfere destructively.

The slits are typically a fraction of a millimeter apart, still many times the wavelength of light, so they send nearly parallel rays to any point on the screen off the axis of symmetry. The two wavelets leave their slits in phase, but one has farther to travel. If that **path difference** is a whole-number multiple of the wavelength, a bright band appears. If it is an odd number of half wavelengths, the waves cancel and a dark band appears.

Part 4 — Total Internal Reflection and Fiber Optics

When a ray passes into a medium of lower optical density, it bends away from the normal, as long as the angle Snell's Law predicts is no more than 90°. If Snell's Law predicts an angle whose sine is greater than 1, which no real angle has, the ray cannot escape and is reflected back into the denser medium. For any such boundary there is a **critical angle**: any ray in the denser medium striking the boundary beyond it is **totally internally reflected**.

Optical fibers put this to work. A fiber is a thin, solid strand of very pure glass or plastic: a **core** surrounded by a **cladding** of slightly lower optical density. Laser light entering the core at the right angle bounces along by total internal reflection at the core–cladding boundary, traveling long distances with minimal loss.

Fiber optics carry much more information than copper wires, are immune to electromagnetic interference, and their signals travel much farther before they need boosting. That makes them ideal for high-speed internet, telecommunications, and medical imaging tools such as endoscopes.

Part 5 — Color Vision

As Episode 1 established, the colors we see are **biological artifacts**, not objective qualities of light. In dim light, vision is essentially black and white, really shades of gray, produced by the **rods** in the retina. In normal daylight we see in color, which adds an enormous amount of vital information about our surroundings.

Color comes from three kinds of **cones**, named red, green, and blue, or long-, medium-, and short-wavelength cones. Each responds to a broad range of frequencies centered on a peak, not to a single color. Red cones peak around **565 nm** and are the only cones still responding well out near the red end of the spectrum, around 700 nm. Blue cones peak around **430 nm**, near the violet end at about 400 nm. Green cones peak around **535 nm**, mid-spectrum. The ranges overlap considerably, and the brain compares the three signals to produce the colors we perceive.

Nearly every color we see can be reproduced by mixing intensities of the **primary colors** pure red, green, and blue. In **additive color theory**, full-intensity red, green, and blue make **white**; none of them makes **black**. Red plus green makes **yellow**, the complement of blue; green plus blue makes **cyan**, the complement of red; red plus blue makes **magenta**, the complement of green. Each pair of **complementary colors** adds up to white.

† "Red" cones peak at about 565 nm, which is actually yellow-green light. They earn the name because they are the only cones still responding strongly near the red end of the spectrum.

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Part 6 — The Constant Speed of Light

The speed of light in a vacuum, **c**, is the **same for every observer**, no matter how the source or the observer is moving, and it is astoundingly large: roughly **3 × 10⁸ meters per second**. Light takes only about 1.3 seconds to reach Earth from the Moon. At that speed you could circle Earth about 7.5 times in one second, if you could survive the accelerations, and reach the Sun in about 500 seconds, or 8.3 minutes.

In an optical or transparent medium, light slows down, sometimes considerably. The **index of refraction** of a medium is the speed of light in a vacuum divided by its speed in the medium: **n = c ÷ v**. Water's index is **1.33**, so light travels at about three-fourths of c in water.

Material particles can never reach or exceed c. But charged particles such as electrons *can* move faster than light itself travels in a medium like water, while still slower than c. When they do, they give off **Cherenkov radiation**, a distinctive blue glow, akin to the sonic boom of a plane outrunning sound. It is seen in the water pools around research reactor cores and spent nuclear fuel. It does not contradict the constancy of c: the particles exceed only the speed of light in the medium, never the speed of light in a vacuum.

Part 7 — Light in Astronomy

Almost everything we know about the Universe comes from studying light from glowing objects and collections of glowing objects. Nearby planets and moons shine by reflected sunlight, while stars glow with their own light across a wide range of frequencies. Farther out we see collections of stars, or galaxies, and farther still, collections of galaxies.

The light may be visible or invisible. Because Earth's atmosphere blocks much of the invisible light, **space telescopes** collect infrared, ultraviolet, X-ray, and gamma-ray light, while **radio telescopes** on the ground collect radio waves.

Because light's speed is finite, about **186,000 miles per second**, looking farther away means **looking farther back in time**. The **Hubble Space Telescope** has seen galaxies whose light left them more than 13 billion years ago. The **James Webb Space Telescope**, built to see in the infrared, looks back even farther, to galaxies that existed only a few hundred million years after the Big Bang, about 13.5 billion years ago, close to the era of the very first stars.

Part 8 — Measuring the Speed of Light

Galileo tried to time light with lanterns on distant hilltops, but light was far too fast. The first successful measurement came in **1676** from the Danish astronomer **Ole Rømer**, who timed the eclipses of Jupiter's moon **Io** and noticed they came early or late depending on how far Earth was from Jupiter. From his work came an estimate of about **220,000 km/s**, short of the current value of **299,792.458 km/s** but remarkable for its time.

This speed matters in astronomy precisely because it is a constant; the value of the speed of light in a vacuum is a **law of physics**. Around **1728**, the English astronomer **James Bradley** used **stellar aberration**, the slight shift in the apparent positions of stars caused by Earth's motion around the Sun, to find about **301,000 km/s**.

In the 19th century, a trio of scientists measured light's speed right here on Earth. In **1849**, **Hippolyte Fizeau** used a rotating toothed wheel and a distant mirror to find about **313,000 km/s** in air. In **1862**, **Léon Foucault** used a rapidly rotating mirror and a fixed mirror to get about **298,000 km/s**. In **1879**, **Albert Michelson** refined Foucault's method to get about

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299,910 km/s, and in **1926** he obtained **299,796 km/s**, a near-perfect result.

Today, **Lunar Laser Ranging** sends coherent light from powerful lasers to the Moon to measure its distance. **Retroreflectors**, arrays of corner-cube prisms, send the light straight back. Five have been used for decades: three left by the Apollo 11, 14, and 15 astronauts and two carried by the Soviet Union's robotic Lunokhod rovers; more recent landers have added newer ones. We know the Moon's distance to millimeter precision, and that it is moving away from Earth by about **3.8 centimeters per year**, a precision possible only because we understand light and its constant speed.

From a mirror to the edge of the observable Universe, every phenomenon in this episode comes down to how light behaves when it meets matter, and to the one speed it never exceeds.

† Rømer himself did not publish a speed. The estimate of about 220,000 km/s was calculated by Christiaan Huygens from Rømer's finding that light takes about 22 minutes to cross Earth's orbit.

TRY IT YOURSELF

Slow light, fast Moon. Diamond has an index of refraction of about 2.4. Use n = c ÷ v to find how fast light travels inside a diamond. Then: a laser pulse fired at an Apollo retroreflector returns to Earth 2.56 seconds after it was sent. Taking c as 300,000 km/s, about how far away is the Moon?

(Answer: v = c ÷ n = 300,000 ÷ 2.4 = 125,000 km/s, well under half of c. For the Moon, 2.56 s is the round trip, so the one-way time is 1.28 s, and the distance is 300,000 × 1.28 = 384,000 km. That matches the "about 1.3 seconds" from the Moon in the episode.)

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STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Reflection

1. What is specular reflection, and what does “image integrity” mean? Can a reflection with image integrity still be distorted?

2. What is diffuse reflection, and what kind of surface produces it?

3. State the Law of Reflection. What is the normal?

4. Both mirrors and rough surfaces obey the Law of Reflection. So why does a mirror form an image while a sheet of paper does not?

5. A ray strikes a flat mirror at 35° from the normal. At what angle from the normal is it reflected?

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Part 2 — Refraction

1. What causes refraction?

2. State Huygens' Principle. How do Huygens' Principle and the Principle of Superposition together explain why a wave front bends when it enters a new medium?

3. Light passes from air into glass, a denser optical medium. Does it speed up or slow down, and does the ray bend toward or away from the normal? What happens when it leaves the glass?

4. What law describes refraction mathematically?

5. Describe the images that the corneas and lenses of healthy eyes form on the retina, and give an everyday optical illusion caused by refraction.

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Part 3 — Diffraction and Young's Experiment

1. What is diffraction? How does it differ from refraction?

2. Why couldn't light diffract if it were made of tiny classical particles, as Newton believed? What kind of light is ideal for demonstrating diffraction?

3. Describe Thomas Young's experiment of about 1801 and the pattern it produced. Why was it important?

4. What are maxima and minima in an interference pattern?

5. What is the path difference? What path difference produces a bright band, and what produces a dark band?

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Part 4 — Total Internal Reflection and Fiber Optics

1. Under what conditions does total internal reflection happen? What is the critical angle?

2. Can total internal reflection happen when light goes from air into water? Explain.

3. Describe the structure of an optical fiber and how it keeps light inside.

4. Give three advantages of fiber optics over copper wires, and two of their uses.

Part 5 — Color Vision

1. Which retinal cells give us vision in dim light, and what does that vision look like?

2. Name the three kinds of cones, both ways, and give the approximate peak sensitivity of each.

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3. Why is it important that each kind of cone responds to a broad range of frequencies rather than a single one? How does the brain produce the colors we perceive?

4. In additive color theory, what combinations of red, green, and blue light make white, black, yellow, cyan, and magenta?

5. What are complementary colors? Name the complement of blue, of red, and of green.

Part 6 — The Constant Speed of Light

1. What is c, and what is special about it? Give its approximate value.

2. About how long does light take to reach Earth from the Moon and from the Sun?

3. Define the index of refraction. Water's index is 1.33. About how fast does light travel in water?

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4. What is Cherenkov radiation, what does it look like, and where is it commonly seen?

5. Explain why Cherenkov radiation does not contradict the rule that nothing material can reach or exceed c.

Part 7 — Light in Astronomy

1. How do we see nearby planets and moons, and how do we see stars?

2. Why do astronomers put infrared, ultraviolet, X-ray, and gamma-ray telescopes in space, while radio telescopes stay on the ground?

3. Why does looking at more distant objects mean looking farther back in time?

4. Compare how far back in time the Hubble and James Webb Space Telescopes have seen. Why can Webb see farther back?

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Part 8 — Measuring the Speed of Light

1. Why did Galileo's lantern experiment fail?

2. How did Ole Rømer's 1676 observations of Io lead to the first estimate of light's speed?

3. What is stellar aberration, and what value for c did James Bradley get from it?

4. Name the trio who measured light's speed on Earth in the 19th century, with the method and approximate result of each.

5. What is Lunar Laser Ranging? What returns the light, and what has it revealed about the Moon?

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YOUR NOTES

Notes

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